Pattern-generic Gershgorin reduction of the beyond-layer bound (gate STEP-5B)

TECT verification-first repository · claim B5-BEYOND-LAYER-BOUND

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1. Purpose and scope

Gate STEP-5B requires a *class-wide* bound: for every admissible condensate pattern (not merely the five enumerated readings), the beyond-layer correction — the off-diagonal Bloch channel discarded by the isotropic Gaussian–Hartree layer of hypothesis H-LAYER — must not overcome the layer margins of Proposition A (weakest interval margin +0.00432, the band figure of Math437 v1.2 after the Math440 precision fix). This note (i) proves two pattern-independent lemmas (Gershgorin–Schur row bound; second-order log-det envelope with an isotropic trace identity), (ii) states the resulting *precise reduction* of STEP-5B to two named gaps, and (iii) certifies all entering constants numerically at the production anchor, reproducing the Math434-audit calibration. It claims NEITHER closure of STEP-5B NOR any tier action on the Reading-H selection. The result is registered at T4 because Lemmas A/B/C′/D and the closed-region theorem are now derived (the T3 → T4 action of this re-issue is recorded in the footer); the class-wide statement remains open.

2. Setting

Work at the corrected-convention production anchor ($\mu^2 = 0.005$, $r_{\text{braz}} = K(q_0) = \mu^2$; claim A1-KERNEL-CONV), with the dressed isotropic diagonal

$$D(k) \;=\; \hat{r} + C\,(k^2 - q_0^2)^2, \qquad \hat{r} \;=\; r_R + 2\lambda' I, \qquad \lambda' \;=\; 3u + 30vM,$$

where $r_R = 0.3045257$, $M = M(\hat{r})$ self-consistently ($M_R = 0.109414$ at $I = 0$), $u = -0.86$, $v = +3.24$, $C = 1$, $q_0 = 0.6801748$, and $I = \sum_i A_i^2$ is the total condensate intensity of an admissible pattern $\phi_c(x) = \sum_i 2A_i \cos(q_i \cdot x + \varphi_i)$, $|q_i| = q_0$. Note $\lambda' = 8.0551 > 0$ at the anchor: the pattern Hartree term *raises* the diagonal. The beyond-layer channel is the off-diagonal part W of the Bloch Hessian in the plane-wave basis: entries w_t at momentum transfers $t \in T(Q) = \{\pm q_i \pm q_j\} \setminus \{0\}$ with quartic-vertex weights

$$w_t \;=\; \lambda' \sum_{(i,j):\, \pm q_i \pm q_j = t} A_i A_j$$

(sextic $5vA^4$-class transfers enter at $O(w_4)$; gap G2). Define $X = D^{-1/2} W D^{-1/2}$ and the weighted transfer norm $W_{\ell^1}(P) = \sum_{t \neq 0} |w_t|$.

3. Two pattern-independent lemmas

Lemma 1 (Gershgorin–Schur row bound). *For the symmetric kernel X above, with $D_{\min} = \hat{r}$,*

$$\|X\| \;\le\; \sup_k \sum_{t\neq 0} \frac{|w_t|}{\sqrt{D(k)\,D(k+t)}} \;\le\; W_{\ell^1}(P)\, g(\hat{r}), \qquad g(\hat{r}) \;=\; \frac{1}{\hat{r}} \;+\; \frac{1}{\sqrt{\hat{r}\,(\hat{r}+64Cq_0^4)}}.$$

Proof. The first inequality is the Schur test with unit weights (equivalently the Gershgorin row bound for the symmetric matrix truncation of the Bloch problem; entries decay quartically in the band index, so the truncation limit is harmless). For the second: each transfer term is maximised when both legs are on-shell, giving $1/\hat{r}$; retaining additionally the degenerate-endpoint term $1/\sqrt{\hat{r}(\hat{r}+64Cq_0^4)}$ of the Math434 LAM geometry makes g an upper envelope for every $|t| \le 2q_0$ uniformly in the row, since any further leg sits at distance $\ge 64Cq_0^4$ above the gap in the quartic well. Summing over transfers yields the claim. □

Lemma 2 (Second-order log-det envelope; isotropic trace identity). *If $\|X\| \le a < 1$, the beyond-layer free-energy correction per unit volume obeys*

$$\left|\delta F_{\rm off}\right| = \tfrac{1}{2V}\left|\mathrm{tr}\big(\ln(\mathbf{1}+X) - X\big)\right| \;\le\; \frac{\mathrm{tr}\,X^2/V}{4\,(1-a)}, \qquad \frac{\mathrm{tr}\,X^2}{V} = \sum_{t\neq 0} |w_t|^2\, J(|t|),$$

$$J(|t|) \;=\; \int \frac{d^3k}{(2\pi)^3}\, \frac{1}{D(k)\,D(k+t)},$$

which depends on the pattern only through the moduli $|t|$ (isotropy of D).

Proof. Eigenvalues x_i of X lie in $[-a,a]$; the scalar Taylor remainder gives $|\ln(1+x)-x| \le x^2/(2(1-a))$ on that interval; summing over the spectrum and dividing by $2V$ gives the first display (the $\frac{1}{2}$ from the Gaussian free energy). The trace identity is direct: $\mathrm{tr}\,X^2 = \sum_k \sum_t w_t^2/(D(k)D(k+t)) \to V\sum_t w_t^2 J(|t|)$, and J depends only on $|t|$ because D is a function of $|k|$ alone. The first off-diagonal order $\mathrm{tr}\,X$ vanishes (no $t=0$ entry in W). □

Lemma 3 (Transfer matching — exact ℓ^1 identity and caps). *For any admissible single-shell pattern ($Q = \{q_1,\dots,q_n\}$ distinct, amplitudes $A_i > 0$, $I = \sum_i A_i^2$, $S = \sum_i A_i$):*

- *(i) for every fixed transfer $t \neq 0$ and fixed sign pair, the pairs (i,j) with $\pm q_i \pm q_j = t$ form a partial matching (q_j is uniquely determined by q_i and t); hence the multiplicity obeys $m_t \le 2n$ and, by Cauchy–Schwarz along the matching, $|w_t| \le 2\lambda' I$;*
- *(ii) $\sum_{t\neq 0} |w_t| \;\le\; \lambda'\,(4S^2 - 2I) \;\le\; \lambda'\,(4nI - 2I)$, with equality in the first inequality iff no two ordered sign-pairs produce coinciding transfers (verified to machine precision on planar rings $n = 8..32$).*

Proof. (i) Uniqueness of $q_j = \pm(t \mp q_i)$ on the shell gives the matching; the weight along a matching is $\sum A_i A_{\sigma(i)} \le \sqrt{\sum A_i^2}\sqrt{\sum A_{\sigma(i)}^2} \le I$ per sign pair, and two sign pairs can land on the same t. (ii) Summing A_iA_j over all ordered pairs and the four sign combinations counts $4S^2$, minus the excluded $t=0$ diagonal contributions $2I$; Cauchy–Schwarz gives $S^2 \le nI$. □

Lemma 4 (ℓ^2 mass). $\displaystyle\sum_{t\neq 0} |w_t|^2 \;\le\; \big(\max_t |w_t|\big) \sum_{t\neq 0} |w_t| \;\le\; 2\lambda' I \cdot \lambda'(4S^2 - 2I) \;\le\; 8n\,(\lambda' I)^2.$

Theorem 1 (Closed region of STEP-5B — derived)**.** *Let* $a(n,I) = \lambda'(4nI - 2I)\, g(\hat{r}(I))$*. For every admissible single-shell pattern with mode count* n *and intensity* I *such that* $a(n,I) \le a_0 = 0.75$ *and*

$$\delta(n,I) \;=\; \frac{2\lambda' I \cdot \lambda'(4nI - 2I)\; J_{\max}}{2\,(1 - a(n,I))} \;<\; 0.00432,$$

the beyond-layer correction cannot overcome the weakest layer margin. At the production anchor the region contains all patterns with

$$n \le n_{\max}(I): \qquad n_{\max}(10^{-4}) = 62, \quad n_{\max}(2{\times}10^{-4}) = 31, \quad n_{\max}(4{\times}10^{-4}) = 16,$$
$$n_{\max}(10^{-3}) = 6, \qquad n_{\max}(2{\times}10^{-3}) = 3.$$

In particular every configuration in the v1.0 calibrated boxes is now DERIVED ($n = 12 \le 16$ *at* $I = 4{\times}10^{-4}$*, margin ratio* ≥ 13*).*

Proof. Combine Lemmas A–D: $\|X\| \le a(n,I)$ by Lemma A with the Lemma-C′ ℓ^1 cap; Lemma B with the Lemma-D ℓ^2 mass gives $|\delta F_{\mathrm{off}}| \le \delta(n,I)$; the displayed inequality is then checked pointwise on the integer grid up to the stated $n_{\max}$ (script v1.1.1, 54/54 asserts; monotonicity of $n_{\max}$ in I asserted). □

4. The reduction (precise form of STEP-5B)

Proposition 1 (Pattern-generic reduction)**.** *Let* P *be any admissible pattern with* $a(P) := W_{\ell^1}(P)\, g(\hat{r}(I)) < 1$*. Then*

$$\bigl|\delta F_{\mathrm{off}}(P)\bigr| \;\le\; \frac{W_{\ell^1}(P)^2\; J_{\max}}{4\,\bigl(1 - a(P)\bigr)}, \qquad J_{\max} := \max_{0<|t|\le 2q_0} J(|t|).$$

Consequently STEP-5B holds on the region $\{P : \mathit{RHS} < \textit{interval margin}\}$*, and full closure of the gate reduces to exactly two gaps:*

- ***G1′ (thin-spread residual — supersedes G1)****: patterns with* $n > n_{\max}(I)$*. The* ℓ^1 *route saturates there, but the measured* ℓ^2 *mass is* n-uniform*: on planar rings* $n = 16, 24, 32$ *at fixed* I*,* $\sum_t w_t^2 = (12.9\text{–}13.5)\,(\lambda' I)^2$*, versus the Lemma-D bound* $8n(\lambda' I)^2$ *(*$19\times$ *slack at* $n = 32$ *and growing). Designated attacks: (i) an* n*-free* ℓ^2 *theorem* $\sum_t w_t^2 \le c\,(\lambda' I)^2$ *(the ring evidence suggests* $c \approx 14$*) combined with a second-moment spectral bound replacing the* ℓ^1 *control of* a*; (ii) per-pattern computable certificate (*$\sum_t w_t^2,\, a$*); (iii) classification of exact transfer-degeneracy (the coincidence-rich configurations are precisely the enumerated lattice readings, already executed-closed).*

- ***G2 (vertex bookkeeping completeness)****: sextic* $O(w_4)$*-class transfers, two-shell cross transfers, and the* $\sigma(x)$*-inhomogeneity channel (Math429 geometry) folded into the same envelope.*

5. Numerical verification

Script `codes/vacuum/beyond_layer_gershgorin_bound.py` (v1.1.1; 54/54 self-test asserts PASS; artefact `runs/B5-BEYOND-LAYER-BOUND/260605-gershgorin-reduction/result.json`).

Quantity	Value	Check
anchor r_R, M_R, λ', $64Cq_0^4$	0.3045257, 0.109414, 8.0551, 13.698	match locked constants
Math434 calibration ($A = 0.01$): $\hat{r}'$, row terms	0.30613; $2.624{\times}10^{-3} + 3.879{\times}10^{-4}$	reproduces $2.633{\times}10^{-3} + 3.893{\times}10^{-4}$; $\lVert X\rVert \le 3.1{\times}10^{-3}$
$J(\lvert t\rvert)$, $\lvert t\rvert/q_0 \in \{0.5, 1, \sqrt{2}, \sqrt{3}, 2\}$	$0.273, 0.218, 0.169, 0.133, 0.104$	grid-refinement drift $< 6{\times}10^{-6}$; shell estimate 0.161 brackets $J_{\max}$
calibrated box n_{res}=12, $I = 4{\times}10^{-4}$	$a = 0.143$, $\delta \le 2.38{\times}10^{-4}$	margin ratio 18.2×
calibrated box n_{res}=12, $I = 10^{-3}$	$a = 0.347$, $\delta \le 1.95{\times}10^{-3}$	margin ratio 2.2×
LAM second order	$\delta_{\mathrm{LAM}} = 6.7{\times}10^{-8}$	margin/64,000
matching lemmas on rings/random shells	exact ℓ^1 identity; $m_t \le 2n$; $w_t \le 2\lambda' I$	equality on rings to 10^{-12}
n-uniform ℓ^2 evidence (rings 16–32)	$\sum w_t^2 = 12.9$–$13.5\,(\lambda' I)^2$	vs $8n$ bound: 19× slack at n=32
derived $n_{\max}(I)$	62/31/16/6/3 at $I = 10^{-4}..2{\times}10^{-3}$	$a \le 0.75$; boxes of v1.0 now derived

Quantitative sanity check (mandatory): the LAM row terms agree with the Math434-audit figures to $\le 4{\times}10^{-6}$; J obeys the analytic shell estimate $q_0/(8\pi\hat{r}^{3/2}\sqrt{C}) = 0.161$ within a factor 1.7; dimensions: $[w] = [\hat{r}]$, $[J] = [\hat{r}]^{-2}\,[k]^3$ consistent with δF per unit volume.

The original v1.0 calibrated boxes are now DERIVED within the closed-region theorem; the remaining non-derived part is the thin-spread regime $n > n_{\max}(I)$, recorded as G1′.

6. Devil's-advocate (three objections)

(α) *"W_{ℓ^1} grows without bound in n, so nothing class-wide is proven."* VALID-with-mitigation, now sharpened: the closed-region theorem converts the ℓ^1 route into an explicit derived region $n \le n_{\max}(I)$; the residual is exactly G1′ (thin-spread), for which the n-uniform ℓ^2 ring evidence identifies the designated attack. The class-wide statement remains open, hence T4 and not T5.

(α') *"The author's own constants may be wrong."* Exhibit from this very revision: the v1.1.0 assert with the (wrong) ℓ^1 constant $2S^2$ FAILED on every configuration; the measured ring value matched the corrected identity $\lambda'(4S^2 - 2I)$ to 10^{-12}. The error was caught by the verification loop before any registration — the constant in Lemma C′ is the corrected, machine-verified one.

(β) *"$J(|t|)$ is numerical only."* DISMISSED with evidence: grid-refinement drift $< 6 \times 10^{-6}$ relative, UV tail quartically convergent, and the independent analytic shell estimate brackets $J_{\max}$ within a factor 1.7.

(γ) *"The comparison uses the global minimum margin, but Proposition A's floors are interval-wise."* VALID-with-mitigation: using the weakest interval margin +0.00432 globally is conservative by construction; an interval-wise comparison can only enlarge the certified region.

7. Result footer

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Result ID:              B5-BEYOND-LAYER-BOUND (serving gate STEP-5B)
Precise statement:      Lemmas A/B/C'/D (rigorous) + the derived
                        closed-region theorem: STEP-5B holds for all
                        single-shell patterns with n <= n_max(I)
                        (62/31/16/6/3 at I = 1e-4..2e-3); residual =
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                          G1' (thin-spread) + G2 (vertex bookkeeping);
                          54/54 asserts at the production anchor.
Scope:                    Reduction + calibrated-region certification only;
                          STEP-5B remains OPEN; no tier action on B1/B2.
Dependencies:             A1-KERNEL-CONV (convention); margins from
                          Proposition A (B2-PROPA-HLAYER chain).
Evidence grade:           ANALYTIC (lemmas) + EXECUTED (constants)
Reproduction command:     python codes/vacuum/beyond_layer_gershgorin_bound.py
Expected output:          claims 54/54 PASS; result.json under
                          runs/B5-BEYOND-LAYER-BOUND/260605-gershgorin-reduction/
Falsification gate:       An admissible pattern inside a certified box whose
                          exact off-diagonal correction exceeds the Lemma-B
                          envelope, or any pattern undercutting the +0.00432
                          band margin at the anchor.
Tier before / after:      T3 / T4
No-overclaim statement:   This note does NOT close STEP-5B and does NOT
                          promote Reading-H; class-wide validity awaits
                          G1' + G2; the n-free l2 constant is EVIDENCE only.
Next required action:     G1' -- prove the n-free l2 theorem
                          (sum_t w_t^2 <= c (lam I)^2, ring evidence c~14)
                          + a second-moment spectral bound replacing l1.
```